# SAMARA OGMS, Russian Federation

WMO: **289000**

Lat: **53.2480N** Lon: **50.2069E** Elev: **139** StdP: **99.66** Time Zone: **4.00 (E04)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **2** | **-26.2** | **-23.1** | **-29.8** | **0.2** | **-26.3** | **-26.2** | **0.4** | **-22.6** | **12.6** | **-6.1** | **11.2** | **-5.3** | **2.8** | **350** | **0.620** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **10.6** | **32.9** | **19.5** | **30.9** | **19.0** | **29.0** | **18.3** | **21.3** | **28.1** | **20.5** | **27.4** | **19.6** | **26.3** | **3.8** | **120** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **19.1** | **14.2** | **23.8** | **18.1** | **13.3** | **22.6** | **17.1** | **12.4** | **22.2** | **62.6** | **28.0** | **59.5** | **27.3** | **56.7** | **25.9** | **24.6** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **10.3** | **9.1** | **8.1** | DB | **-29.0** | **35.4** | **3.4** | **1.9** | **-31.4** | **36.7** | **-33.4** | **37.8** | **-35.3** | **38.9** | **-37.8** | **40.3** | (4) |
| (5) | | | | WB | **-29.2** | **22.8** | **3.4** | **1.3** | **-31.6** | **23.7** | **-33.6** | **24.5** | **-35.5** | **25.2** | **-37.9** | **26.2** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | **Temperatures, Degree-Days and Degree-Hours** | | DBAvg | **5.7** | **-10.3** | **-10.4** | **-3.6** | **6.8** | **14.9** | **19.0** | **21.2** | **19.6** | **13.5** | **6.0** | **-1.8** | **-7.6** | (6) |
| (7) | | | DBStd | **12.65** | **6.88** | **6.72** | **5.02** | **5.45** | **4.93** | **4.48** | **3.59** | **4.35** | **4.24** | **4.57** | **5.65** | **6.44** | (7) |
| (8) | | | HDD10.0 | **2814** | **629** | **571** | **421** | **126** | **13** | **1** | **0** | **0** | **14** | **139** | **353** | **546** | (8) |
| (9) | | | HDD18.3 | **4884** | **887** | **804** | **679** | **346** | **129** | **45** | **14** | **37** | **153** | **382** | **603** | **804** | (9) |
| (10) | | | CDD10.0 | **1245** | **0** | **0** | **0** | **31** | **164** | **271** | **346** | **298** | **119** | **16** | **0** | **0** | (10) |
| (11) | | | CDD18.3 | **275** | **0** | **0** | **0** | **1** | **22** | **65** | **102** | **77** | **9** | **0** | **0** | **0** | (11) |
| (12) | | | CDH23.3 | **2755** | **0** | **0** | **0** | **10** | **259** | **654** | **984** | **758** | **89** | **0** | **0** | **0** | (12) |
| (13) | | | CDH26.7 | **986** | **0** | **0** | **0** | **1** | **64** | **239** | **365** | **300** | **16** | **0** | **0** | **0** | (13) |
| (14) | **Wind** | | WSAvg | **4.0** | **4.2** | **4.2** | **4.3** | **4.2** | **4.1** | **3.8** | **3.4** | **3.6** | **3.6** | **3.9** | **4.1** | **4.2** | (14) |
| (15) | **Precipitation** | | PrecAvg | **495** | **45** | **35** | **31** | **33** | **39** | **48** | **47** | **39** | **44** | **43** | **48** | **42** | (15) |
| (16) | | | PrecMax | **772** | **93** | **71** | **70** | **108** | **102** | **117** | **125** | **108** | **181** | **98** | **136** | **108** | (16) |
| (17) | | | PrecMin | **309** | **3** | **1** | **2** | **0** | **4** | **3** | **0** | **0** | **1** | **2** | **5** | **2** | (17) |
| (18) | | | PrecStd | **116** | **20** | **18** | **16** | **25** | **22** | **33** | **30** | **29** | **41** | **25** | **32** | **22** | (18) |
| (19) | **Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures** | 0.4% | DB | | **2.2** | **2.8** | **10.0** | **24.9** | **30.2** | **34.2** | **35.1** | **35.8** | **28.9** | **20.1** | **11.1** | **3.9** | (19) |
| (20) | | | MCWB | | **1.8** | **1.3** | **6.1** | **13.1** | **17.3** | **18.9** | **19.0** | **19.1** | **16.5** | **11.7** | **9.4** | **3.3** | (20) |
| (21) | | 2% | DB | | **1.1** | **1.2** | **6.2** | **21.2** | **28.6** | **31.9** | **32.2** | **32.2** | **26.0** | **17.2** | **8.8** | **2.1** | (21) |
| (22) | | | MCWB | | **0.7** | **0.7** | **3.4** | **11.5** | **16.3** | **18.7** | **19.7** | **19.2** | **15.8** | **11.1** | **7.4** | **1.5** | (22) |
| (23) | | 5% | DB | | **0.1** | **0.2** | **4.0** | **18.9** | **26.2** | **29.8** | **30.2** | **30.0** | **23.2** | **15.0** | **6.8** | **1.1** | (23) |
| (24) | | | MCWB | | **-0.2** | **-0.2** | **2.0** | **10.6** | **15.4** | **18.4** | **19.4** | **19.0** | **14.6** | **10.1** | **5.8** | **0.7** | (24) |
| (25) | | 10% | DB | | **-1.0** | **-0.9** | **2.3** | **16.1** | **24.0** | **27.1** | **28.8** | **27.8** | **21.0** | **12.9** | **5.0** | **0.2** | (25) |
| (26) | | | MCWB | | **-1.3** | **-1.4** | **1.0** | **9.2** | **14.4** | **17.9** | **18.9** | **18.2** | **13.9** | **9.0** | **4.2** | **-0.1** | (26) |
| (27) | **Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures** | 0.4% | WB | | **2.1** | **1.6** | **6.4** | **14.3** | **19.0** | **21.9** | **22.7** | **21.9** | **18.5** | **14.2** | **10.3** | **3.2** | (27) |
| (28) | | | MCDB | | **2.2** | **2.0** | **9.4** | **21.9** | **27.1** | **29.0** | **29.3** | **29.6** | **24.0** | **17.6** | **11.0** | **3.4** | (28) |
| (29) | | 2% | WB | | **0.6** | **0.7** | **3.6** | **12.7** | **17.5** | **20.6** | **21.6** | **20.8** | **17.0** | **12.1** | **7.3** | **1.6** | (29) |
| (30) | | | MCDB | | **0.8** | **1.0** | **5.7** | **19.6** | **26.1** | **27.9** | **27.7** | **28.5** | **23.5** | **16.1** | **8.0** | **1.8** | (30) |
| (31) | | 5% | WB | | **-0.2** | **-0.1** | **2.2** | **11.0** | **16.5** | **19.6** | **20.8** | **19.7** | **15.8** | **10.6** | **5.6** | **0.6** | (31) |
| (32) | | | MCDB | | **0.0** | **0.2** | **3.5** | **17.9** | **24.2** | **26.8** | **27.1** | **27.1** | **21.1** | **13.7** | **6.3** | **0.9** | (32) |
| (33) | | 10% | WB | | **-1.5** | **-1.4** | **1.2** | **9.7** | **15.3** | **18.6** | **20.0** | **18.7** | **14.6** | **9.6** | **4.1** | **-0.1** | (33) |
| (34) | | | MCDB | | **-1.2** | **-1.0** | **2.1** | **15.3** | **22.1** | **25.2** | **26.2** | **25.7** | **19.9** | **12.3** | **4.7** | **0.1** | (34) |
| (35) | **Mean Daily Temperature Range** | | MDBR | | **6.1** | **7.2** | **7.5** | **9.7** | **11.2** | **10.7** | **10.6** | **10.6** | **9.6** | **7.0** | **4.9** | **5.3** | (35) |
| (36) | | 5% DB | MCDBR | | **5.5** | **5.7** | **7.9** | **13.7** | **14.0** | **13.7** | **13.1** | **13.4** | **13.6** | **11.1** | **6.2** | **4.2** | (36) |
| (37) | | | MCWBR | | **5.3** | **5.2** | **5.7** | **6.9** | **5.9** | **5.2** | **4.8** | **4.9** | **6.3** | **6.4** | **5.4** | **4.0** | (37) |
| (38) | | 5% WB | MCDBR | | **5.5** | **5.5** | **6.8** | **12.8** | **12.5** | **11.7** | **11.2** | **11.8** | **11.6** | **9.4** | **5.5** | **4.1** | (38) |
| (39) | | | MCWBR | | **5.4** | **5.2** | **5.3** | **6.9** | **5.9** | **5.4** | **4.8** | **5.0** | **6.3** | **6.2** | **5.1** | **4.1** | (39) |
| (40) | **Clear-Sky Solar Irradiance** | taub | | | **0.300** | **0.324** | **0.369** | **0.426** | **0.402** | **0.394** | **0.405** | **0.423** | **0.379** | **0.351** | **0.326** | **0.299** | (40) |
| (41) | | taud | | | **2.071** | **2.041** | **2.012** | **2.101** | **2.285** | **2.355** | **2.323** | **2.253** | **2.351** | **2.344** | **2.242** | **2.092** | (41) |
| (42) | | Ebn at Noon | | | **633** | **746** | **791** | **799** | **847** | **857** | **839** | **796** | **790** | **728** | **615** | **554** | (42) |
| (43) | | Edh at Noon | | | **74** | **105** | **134** | **141** | **123** | **116** | **118** | **120** | **96** | **78** | **64** | **60** | (43) |
| (44) | **All-Sky Solar Radiation** | RadAvg | | | **0.91** | **1.97** | **3.35** | **4.34** | **5.63** | **6.08** | **6.01** | **4.91** | **3.24** | **1.64** | **0.70** | **0.50** | (44) |
| (45) | | RadStd | | | **0.15** | **0.20** | **0.38** | **0.44** | **0.41** | **0.64** | **0.50** | **0.40** | **0.45** | **0.28** | **0.11** | **0.14** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (5 neighbors) | **N/A** | **N/A** | **N/A** | **+1.09** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page